PROTEIN STRUCTURE PREDICTION

Title:

Protein Structure Prediction and Structural Analysis of TP53 Using AlphaFold and SWISS-MODEL

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1. Introduction

Proteins are biological macromolecules whose function is highly dependent on their three-dimensional structure. The tumor suppressor protein p53, encoded by the TP53 gene, plays a critical role in regulating the cell cycle, DNA repair, and apoptosis. Mutations or structural disruptions in TP53 are strongly associated with cancer development.

Protein structure prediction enables the computational modeling of a protein's 3D conformation from its amino acid sequence. Tools such as AlphaFold and SWISS-MODEL leverage machine learning and homology modeling techniques to predict biologically relevant structures with high accuracy.

This study aims to predict and analyze the 3D structure of human TP53 protein and identify key structural domains responsible for its biological function.

2. Materials and Methods

2.1 Protein Sequence Retrieval

The amino acid sequence of human TP53 (UniProt ID: P04637) was retrieved from the UniProt database.

2.2 Structure Prediction Tools

Two computational tools were used for structure prediction:

- **AlphaFold Protein Structure Database**
- **SWISS-MODEL (homology modeling platform)**

Both tools provided predicted 3D structural models of TP53 based on its amino acid sequence.

AlphaFold images

Figure 1: AlphaFold Predicted 3D Structure of Human TP53 Protein (P04637)

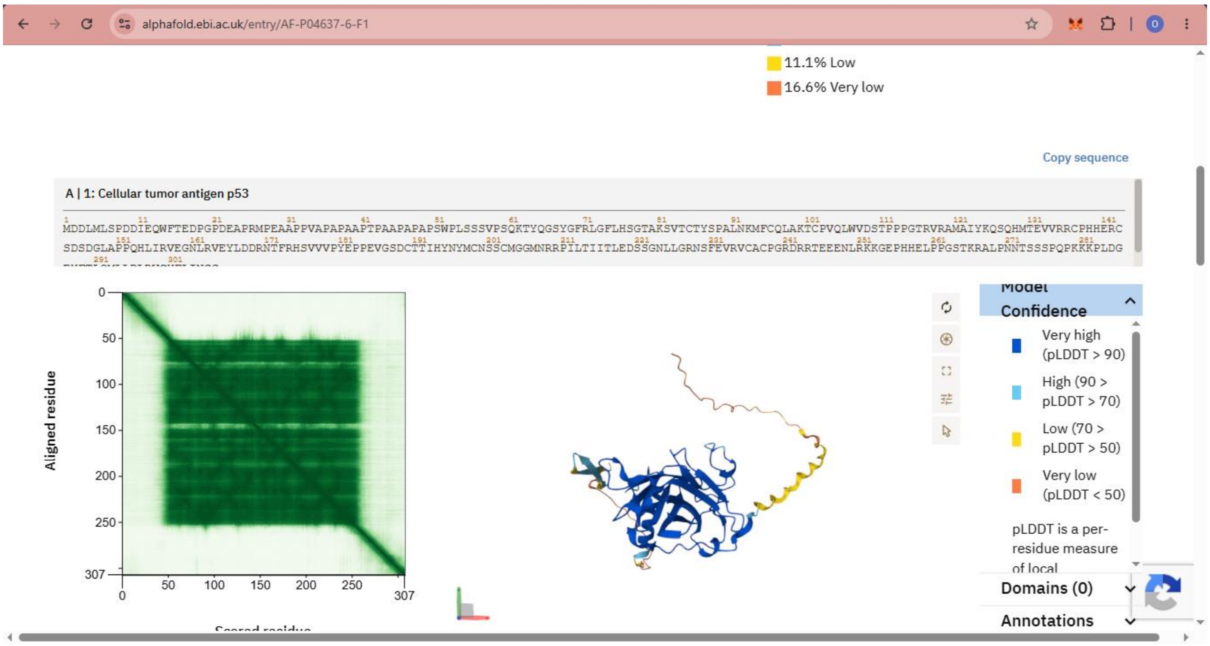


Figure 2: Alternative Orientation of AlphaFold TP53 Protein Structure Highlighting Folded Core Region

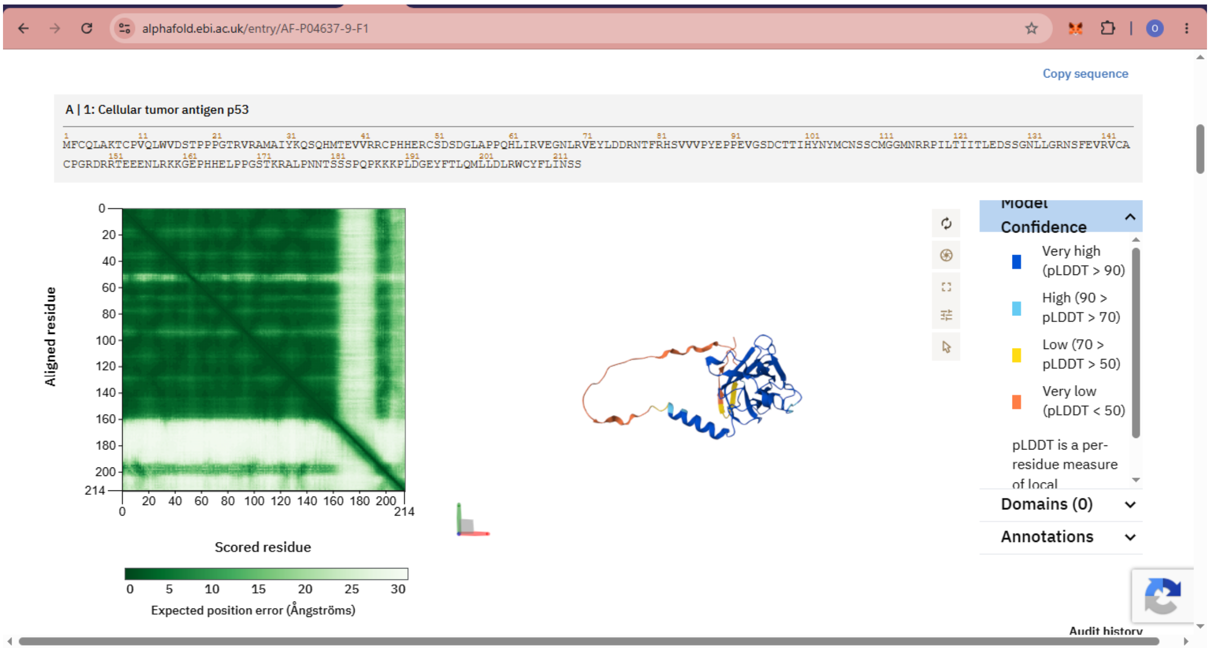
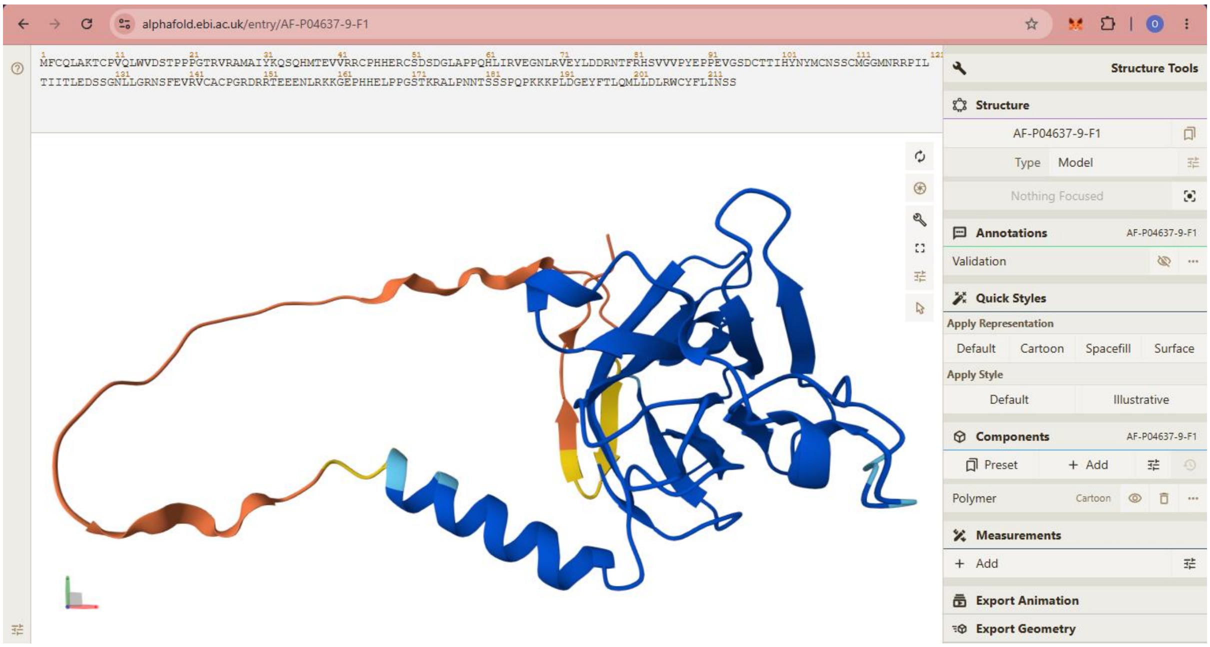
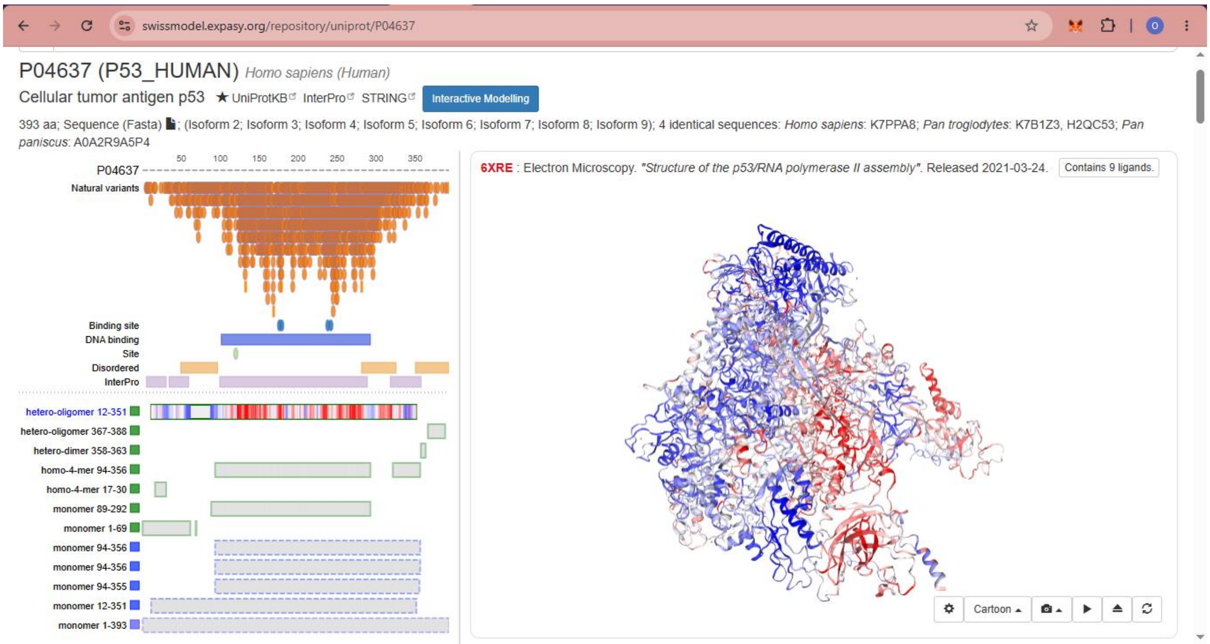


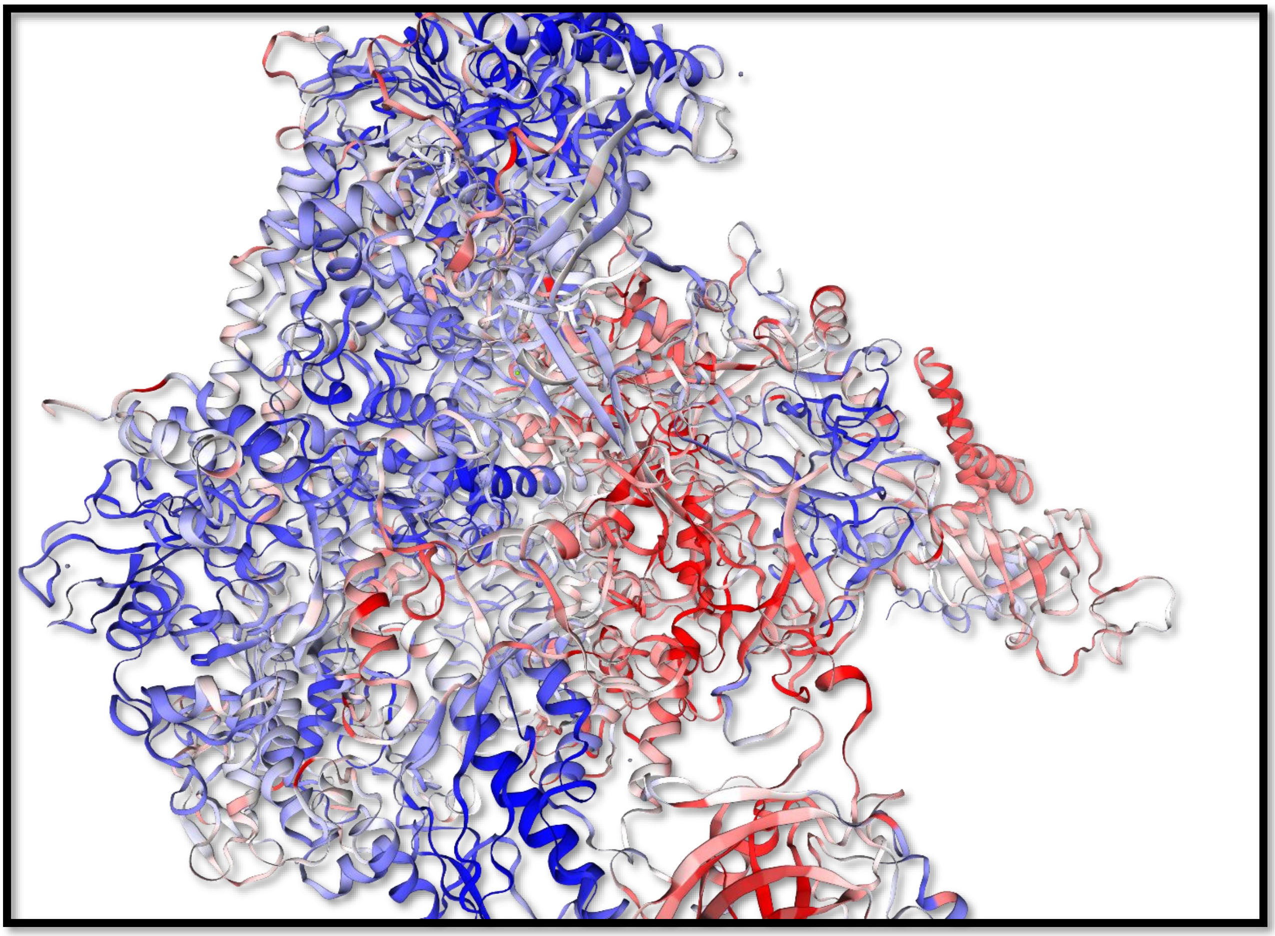
Figure 3: Zoomed View of DNA-Binding Domain in AlphaFold Predicted TP53 Structure



SWISS-MODEL images

Figure 4: SWISS-MODEL Predicted 3D Structure of Human TP53 Protein





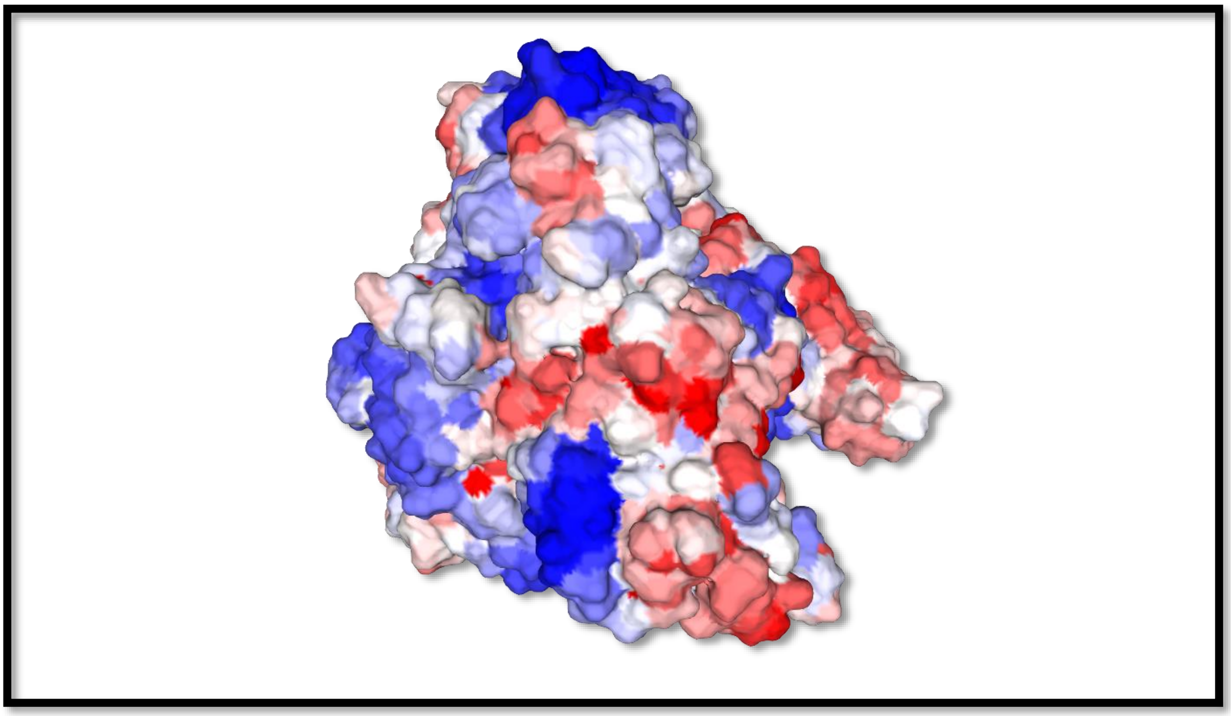
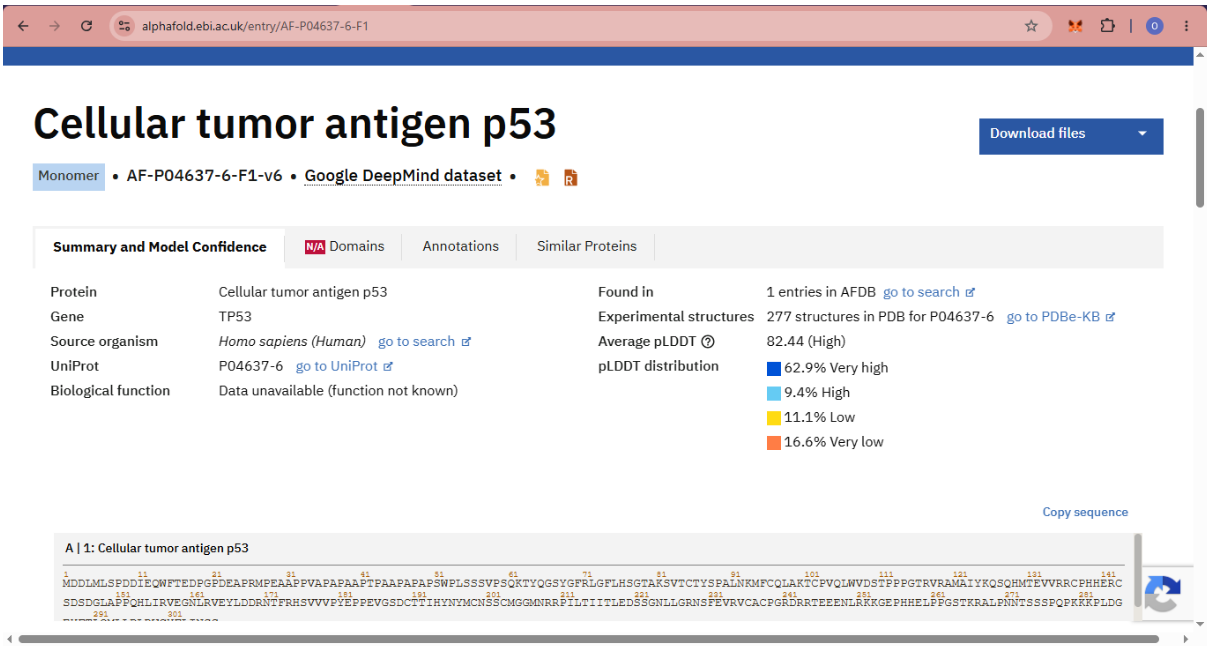


Figure 6: Structural Confidence Representation of TP53 Model from AlphaFold (Color-Coded by Confidence Score)



2.3 Structure Visualization

The predicted structures were visualized using molecular visualization tools:

- PyMOL (desktop software)

Visualization was used to examine secondary structural elements such as alpha-helices, beta-sheets, and loop regions.

2.4 Procedure

1. TP53 protein sequence was obtained from UniProt.
2. The sequence was submitted to AlphaFold and SWISS-MODEL for structure prediction.
3. Predicted models were downloaded or accessed via web interface.
4. Structures were visualized in PyMOL.

5. Structural regions were analyzed, including the DNA-binding domain and terminal regions.
6. Screenshots were captured in different orientations for analysis.

3. Results

The predicted 3D structure of TP53 revealed a complex folded protein consisting of distinct structural domains.

Key observations include:

- A highly structured central core region corresponding to the DNA-binding domain.
- Flexible N-terminal and C-terminal regions exhibiting lower structural rigidity.
- Predominance of alpha-helices and loop regions contributing to protein stability and interaction potential.
- High-confidence structural regions were consistent across both AlphaFold and SWISS-MODEL predictions.

Figure 7: Full 3D Structure of TP53 Protein

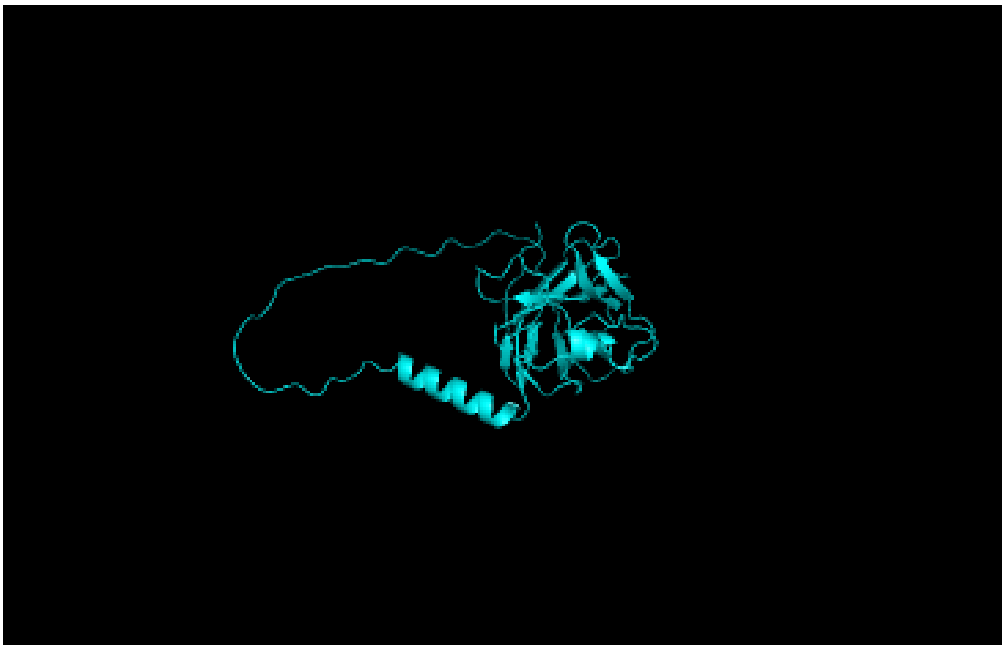


Figure 8: Zoomed DNA-Binding Domain of TP53

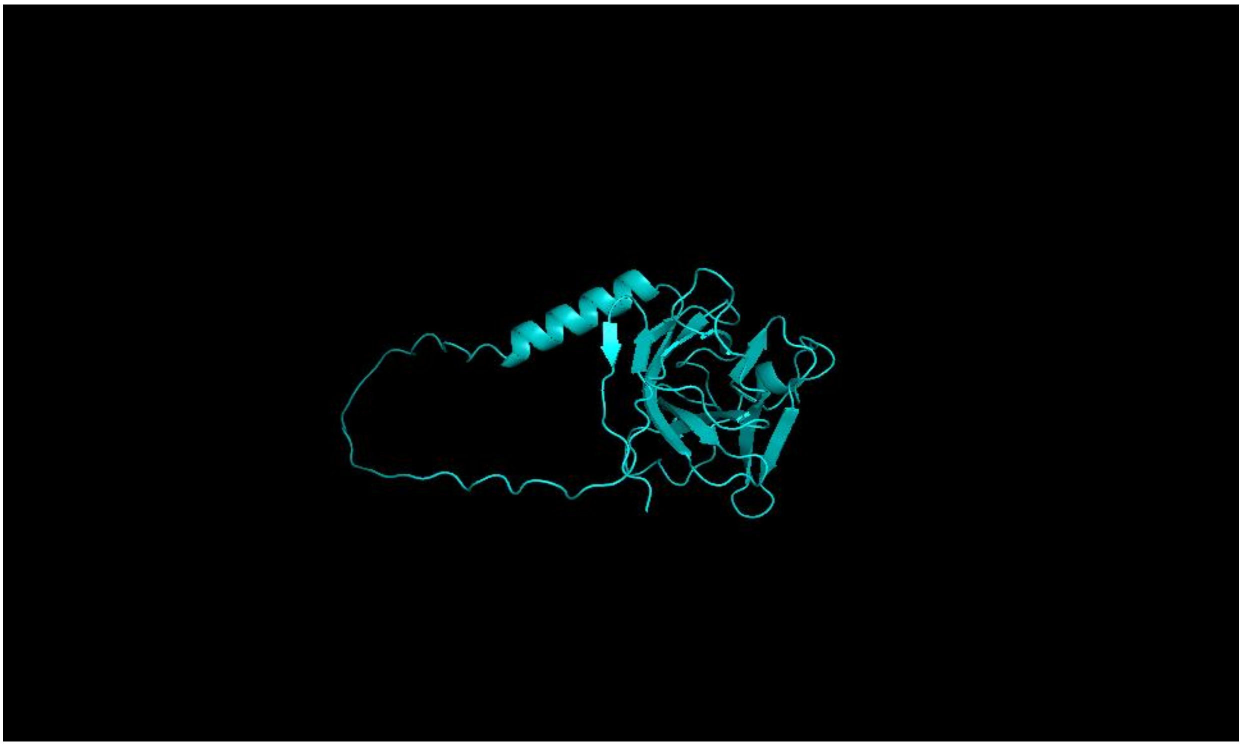
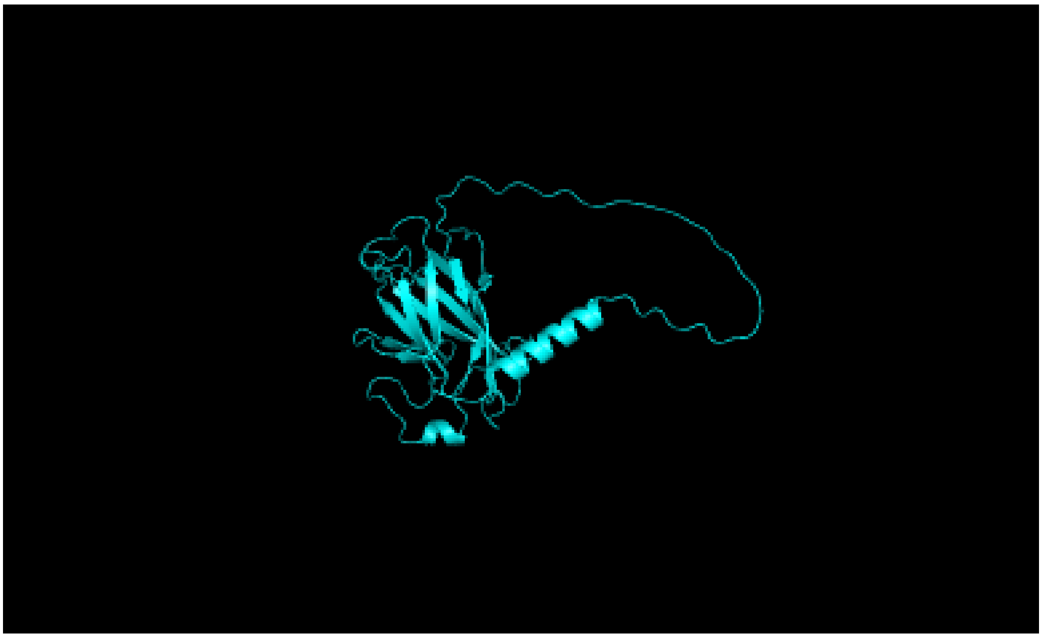


Figure 9: Alternative Orientation of TP53 Structure



4. Interpretation

The structural analysis of TP53 demonstrates a well-defined and highly ordered central domain, corresponding to the DNA-binding region of the protein. This region is critical for TP53 function, as it facilitates binding to DNA sequences involved in cell cycle regulation and apoptosis.

The central core domain appears compact and structurally stable, consistent with its high evolutionary conservation observed in multiple sequence alignment (Task 2). In contrast, the N-terminal and C-terminal regions are more flexible and less structured, indicating potential roles in regulatory interactions and protein-protein binding.

The consistency between AlphaFold and SWISS-MODEL predictions strengthens the reliability of the structural model. Overall, the predicted structure aligns with known biological functions of TP53 as a transcription factor and tumor suppressor.

5. Conclusion

The protein structure prediction of TP53 revealed a highly conserved and structurally stable DNA-binding domain, alongside flexible terminal regions. The results highlight the relationship between protein structure and function, demonstrating that conserved sequences correspond to structurally important regions essential for biological activity.

6. References

- Jumper, J. et al. (2021). *Highly accurate protein structure prediction with AlphaFold.* Nature, 596, 583–589.
- Waterhouse, A. et al. (2018). *SWISS-MODEL: homology modelling of protein structures and complexes.* Nucleic Acids Research, 46(W1), W296–W303.
- The UniProt Consortium (2023). *UniProt: the Universal Protein Knowledgebase.* Nucleic Acids Research.
- PyMOL Molecular Graphics System, Schrödinger, LLC.